

(2) Substitution Method

Problem 1

$$T(n) = \begin{cases} 1 & \text{if } n=1 \\ n * T(n-1) & \text{if } n > 1 \end{cases}$$

Solution

$$T(n) = n * T(n-1) \text{ --- (1)}$$

$$T(n-1) = (n-1) * T(n-1-1)$$

$$= (n-1) * T(n-2) \text{ --- (2)}$$

$$T(n-2) = (n-2) * T(n-3) \text{ --- (3)}$$

$$T(n-3) = (n-3) * T(n-4) \text{ --- (4)}$$

Apply eqn (2), (3), (4) in eqn (1)

$$T(n) = n * T(n-1)$$

$$= n * (n-1) * T(n-2)$$

$$= n * (n-1) * (n-2) * T(n-3)$$

$$= n * (n-1) * (n-2) * (n-3) * T(n-4)$$

So, we can extend the terms till

$$= n * (n-1) * (n-2) * (n-3) * \dots * T(n-(n-1))$$

$$= n * (n-1) * (n-2) * (n-3) * \dots * T(1)$$

$$= n * (n-1) * (n-2) * (n-3) * \dots * T(1)$$

So, it is nothing but $3 * 2 * 1$
 $= n * (n-1) * (n-2) * (n-3) * \dots$ (5)

Simplify eqn (5)

$$= n \times n \left(1 - \frac{1}{n}\right) \times n \left(1 - \frac{2}{n}\right) \times n \left(1 - \frac{3}{n}\right)$$

$$\times \dots \times n \times \frac{3}{n} \times n \times \frac{2}{n} \times n \times \frac{1}{n}$$

Consider n term, which has taken out common

$$= n \times n \times n \times \dots$$

$$= n^n$$

So, the time complexity is $O(n^n)$

Problem 2

$$T(n) = \begin{cases} 1, & \text{if } n = 1 \\ T(n-1) + \log n, & \text{if } n > 1 \end{cases}$$

$$T(n) = T(n-1) + \log n \quad \text{--- (1)}$$

$$T(n-1) = T(n-1-1) + \log(n-1)$$

$$= T(n-2) + \log(n-1) \quad \text{--- (2)}$$

$$T(n-2) = T(n-3) + \log(n-2) \quad \text{--- (3)}$$

Apply Eqn (2) and (3) in Eqn (1)

$$T(n) = T(n-2) + \log(n-1) + \log n$$

$$= T(n-3) + \log(n-2) + \log(n-1) + \log n$$

$$= T(n-4) + \log(n-3) + \log(n-2) + \log(n-1) + \log n \quad \text{--- (4)}$$

Generalize the Equation (4) till k terms

$$= T(n-k) + \log(n-(k-1)) + \log(n-(k-2)) + \dots + \log n \rightarrow (5)$$

$$= T(1) + \log(n-(n-1)) + \log(n-(n-2)) + \dots + \log n$$

$$= T(1) + \log(1) + \log(2) + \log(3) + \dots + \log(n)$$

~~$$= T(1)$$~~

$$= 1 + \log 1 + \log 2 + \log 3 + \dots + \log n$$

$$= 1 + \log(1 \times 2 \times 3 \times \dots \times n)$$

$$= 1 + \log(n!)$$

$$= 1 + \log n^n$$

$$= 1 + n \log n$$

Ignore Constant, so the time

Complexity

is

$$O(n \log n)$$

Consider

$$T(n-k)$$

where $n-k=0$

$$n=k$$

For Time Complexity

Consider

$$n-k=1$$

Apply n and $n-k$ value in eqn (5)